Unit: Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ____________
Date: ________

Quiz (Habanero)

Q1. A heavy rainstorm causes a water level rise of $R = \frac{6}{\sqrt{3}}$ cm/hour. What is this rate in its simplest rationalized form?
(A) $3\sqrt{3}$
(B) $2\sqrt{3}$
(C) $\sqrt{3}$
(D) $6\sqrt{3}$
(E) $\frac{\sqrt{3}}{2}$

Q2. The temperature drop in a mountain ecosystem is modeled by $T = \frac{10}{\sqrt{5}-\sqrt{3}}$ degrees. Simplify the expression.
(A) $5(\sqrt{5}+\sqrt{3})$
(B) $10(\sqrt{5}+\sqrt{3})$
(C) $2(\sqrt{5}+\sqrt{3})$
(D) $5(\sqrt{5}-\sqrt{3})$
(E) $20(\sqrt{5}+\sqrt{3})$

Q3. Soil moisture content M is given by $\frac{\sqrt{2}}{\sqrt{10}}$. Rationalize the denominator to find the decimal equivalent's radical form.
(A) $\frac{\sqrt{20}}{10}$
(B) $\frac{1}{5}$
(C) $\frac{\sqrt{5}}{5}$
(D) $\sqrt{5}$
(E) $\frac{\sqrt{10}}{5}$

Q4. A forest fire spreads over an area $A = \frac{21}{\sqrt{7}}$ square kilometers. What is the rationalized area?
(A) $7\sqrt{3}$
(B) $3\sqrt{7}$
(C) $21\sqrt{7}$
(D) $\sqrt{147}$
(E) 3

Q5. The wind chill factor involves a ratio of $\frac{12}{3-\sqrt{3}}$. Simplify this constant for a weather model.
(A) $6+\sqrt{3}$
(B) $2(3+\sqrt{3})$
(C) $6+6\sqrt{3}$
(D) $4+2\sqrt{3}$
(E) $6-2\sqrt{3}$

Q6. The growth rate of a coral reef is $G = \frac{\sqrt{6}+\sqrt{2}}{\sqrt{2}}$ mm per year. What is the rationalized growth rate?
(A) $\sqrt{3}+1$
(B) $\sqrt{3}+2$
(C) $2\sqrt{3}+1$
(D) $\sqrt{12}+1$
(E) $\sqrt{8}$

Q7. CO2 absorption in a specific wetland is calculated as $\frac{4}{\sqrt{2}+1}$. Which expression is equivalent?
(A) $4\sqrt{2}+4$
(B) $4\sqrt{2}-1$
(C) $4\sqrt{2}-4$
(D) $2\sqrt{2}-2$
(E) $\frac{4\sqrt{2}-4}{3}$

Q8. A glacier recedes at a rate of $V = \frac{15}{3\sqrt{5}}$ meters per month. Simplify V.
(A) $5\sqrt{5}$
(B) $\frac{\sqrt{5}}{3}$
(C) $\sqrt{5}$
(D) $3\sqrt{5}$
(E) $\frac{5\sqrt{5}}{3}$

Open Ended Questions (FRQ)

Q9. The volume of a spherical raindrop is related to its radius by a factor involving $\frac{1}{\sqrt[3]{4}}$. Rationalize the denominator of this expression using a single cube root.

Q10. The efficiency of a solar panel under cloudy conditions is given by the expression $\frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}+\sqrt{2}}$. Rationalize the denominator and simplify the resulting expression completely.

Chef's Tips

- Ensure students multiply by the conjugate when dealing with binomial denominators.
- Remind students that $\sqrt{a}\cdot\sqrt{a}=a$ only for non-negative real numbers.
- Encourage simplifying the fraction before rationalizing to work with smaller numbers.